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(Approved by AICTE, New Delhi, Affiliated to JNTUH, Hyderabad)
AN AUTONOMOUS INSTITUTION
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

II YEAR I SEM **SECTION-B** **A.Y 2025-26**

REAL TIME RESEARCH PROJECT

TITLE OF THE PROJECT: AUTOMATED MINE SAFETY ALERT SYSTEM

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ABSTRACT

Coal mining is one of the most hazardous occupations due to the presence of toxic gases, sudden temperature rise, and underground vibrations that may lead to accidents or mine collapses. Ensuring the safety of workers in underground coal mines is therefore a critical challenge. This project proposes a Coal Mine Safety Monitoring and Alerting System designed to detect dangerous environmental conditions inside mines and provide immediate alerts.

The system is built using an Arduino Uno microcontroller integrated with multiple sensors such as a gas sensor, temperature sensor, and vibration sensor. The gas sensor monitors the presence of harmful gases in underground mines, while the temperature sensor tracks the surrounding temperature to detect abnormal heat levels. The vibration sensor detects unusual ground vibrations that could indicate structural instability.

The collected sensor data is processed by the Arduino controller in real time. The system displays the environmental readings on an LCD display, allowing workers to monitor the current conditions directly. When any parameter exceeds the predefined safe threshold levels, the system activates a buzzer alarm to alert workers about the dangerous situation.

The proposed system is simple, low-cost, and reliable, and it works without requiring WiFi or internet connectivity. It provides continuous monitoring and quick alerts, helping to improve the safety and protection of workers in coal mines.

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